Question ID: 4b76c7f1

Question

$$2x+7y=9$$
$$8x+28y=a$$

In the given system of equations, a is a constant. If the system has infinitely many solutions, what is the value of a ?

Answer

A. 4

B. 9

C. 36

D. 54

Question ID: 44d65912

Question

Angela is playing a video game. In this game, players can score points only by collecting coins and stars. Each coin is worth c points, and each star is worth s points.

- The first time she played, Angela scored 700 points. She collected 20 coins and 10 stars.
- The second time she played, Angela scored 850 points. She collected 25 coins and 12 stars.

Which system of equations can be used to correctly determine the values of c and s ?

Answer

A. $10c + 20s = 700$
$12c + 25s = 850$

B. $20c + 10s = 700$
$25c + 12s = 850$

C. $20c + 700s = 10$
$25c + 850s = 12$

D. $700c + 20s = 10$
$850c + 25s = 12$

Question ID: 17f176ec

Question

A movie theater charges $11 for each full-price ticket and $8.25 for each reduced-price ticket. For one movie showing, the theater sold a total of 214 full-price and reduced-price tickets for $2,145. Which of the following systems of equations could be used to determine the number of full-price tickets, f, and the number of reduced-price tickets, r, sold?

Answer

A. $f+r=2{,}145$
$11f+8.25r=214$

B. $f+r=214$
$11f+8.25r=2{,}145$

C. $f+r=214$
$8.25f+11r=2{,}145$

D. $f+r=2{,}145$
$8.25f+11r=214$

Question ID: 7d89376f

Question

A discount airline sells a certain number of tickets, x, for a flight for $90 each. It sells the number of remaining tickets, y, for $250 each. For a particular flight, the airline sold 120 tickets and collected a total of $27,600 from the sale of those tickets. Which system of equations represents this relationship between x and y ?

Answer

A. $\begin{cases} x+y=120 \\ 90x+250y=27{,}600 \end{cases}$

B. $\begin{cases} x+y=120 \\ 90x+250y=120(27{,}600) \end{cases}$

C. $\begin{cases} x+y=27{,}600 \\ 90x+250y=120(27{,}600) \end{cases}$

D. $\begin{cases} 90x=250y \\ 120x+120y=27{,}600 \end{cases}$

Question ID: 0df106df

Question

An online bookstore sells novels and magazines. Each novel sells for $4, and each magazine sells for $1. If Sadie purchased a total of 11 novels and magazines that have a combined selling price of $20, how many novels did she purchase?

Answer

A. 2

B. 3

C. 4

D. 5

Question ID: 0d1dca87

Question

$$3x + y = 29$$
$$x = 2$$

If (x, y) is the solution to the given system of equations, what is the value of y ?

Question ID: 4fb8adf7

Question

$$4x - 3y = 5$$

$$x = 8$$

What is the solution (x, y) to the given system of equations?

Answer

A. $(8, 9)$

B. $(8, -24)$

C. $(8, -9)$

D. $(8, 24)$

Question ID: cd33b015

Question

$x + y = 20$

$2(x + y) + 3y = 85$

If (x, y) is the solution to the given system of equations, what is the value of y ?

Answer

A. 10

B. 15

C. 60

D. 65

Question ID: b86123af

Question

Hiro and Sofia purchased shirts and pants from a store. The price of each shirt purchased was the same and the price of each pair of pants purchased was the same. Hiro purchased 4 shirts and 2 pairs of pants for $86, and Sofia purchased 3 shirts and 5 pairs of pants for $166. Which of the following systems of linear equations represents the situation, if x represents the price, in dollars, of each shirt and y represents the price, in dollars, of each pair of pants?

Answer

A. $4x + 2y = 86$
$3x + 5y = 166$

B. $4x + 3y = 86$
$2x + 5y = 166$

C. $4x + 2y = 166$
$3x + 5y = 86$

D. $4x + 3y = 166$
$2x + 5y = 86$

Question ID: 608eeb6e

Question

$$5x = 15$$

$$-4x + y = -2$$

The solution to the given system of equations is (x, y). What is the value of $x + y$?

Answer

A. -17

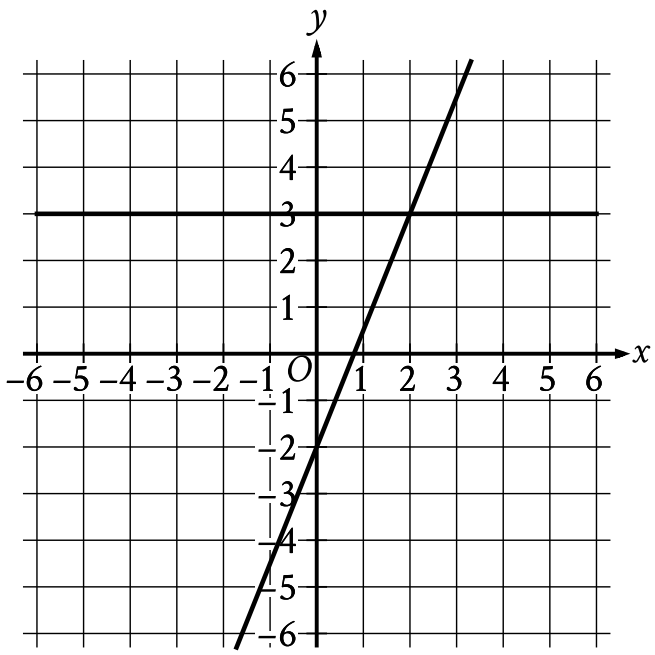
B. -13

C. 13

D. 17

Question ID: b0fc3166

Question



The graph of a system of linear equations is shown. What is the solution (x, y) to the system?

Answer

A. $(0, 3)$

B. $(1, 3)$

C. $(2, 3)$

D. $(3, 3)$

Question ID: dba8d38a

Question

A petting zoo sells two types of tickets. The standard ticket, for admission only, costs $5. The premium ticket, which includes admission and food to give to the animals, costs $12. One Saturday, the petting zoo sold a total of 250 tickets and collected a total of $2,300 from ticket sales. Which of the following systems of equations can be used to find the number of standard tickets, s, and premium tickets, p, sold on that Saturday?

Answer

A. $$\begin{aligned} s+p&=250 \\ 5s+12p&=2{,}300 \end{aligned}$$

B. $$\begin{aligned} s+p&=250 \\ 12s+5p&=2{,}300 \end{aligned}$$

C. $$\begin{aligned} 5s+12p&=250 \\ s+p&=2{,}300 \end{aligned}$$

D. $$\begin{aligned} 12s+5p&=250 \\ s+p&=2{,}300 \end{aligned}$$

Question ID: 8abed0fb

Question

$$y = 2x + 3$$
$$x = 1$$

What is the solution (x, y) to the given system of equations?

Answer

A. $(1, 2)$

B. $(1, 5)$

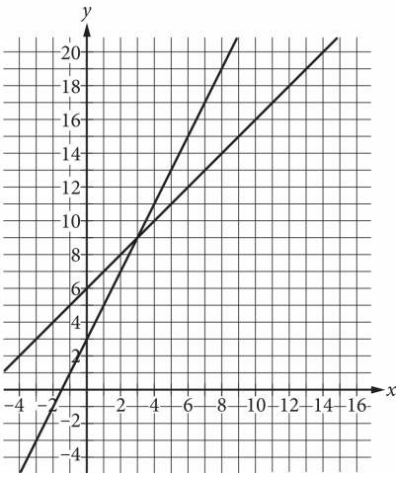
C. $(2, 3)$

D. $(2, 7)$

Question ID: e1259a5a

Question

A system of two linear equations is graphed in the xy-plane below.



Which of the following points is the solution to the system of equations?

Answer

A. $(3,9)$

B. $(6,15)$

C. $(8,10)$

D. $(12,18)$

Question ID: ee031767

Question

A dance teacher ordered outfits for students for a dance recital. Outfits for boys cost $26, and outfits for girls cost $35. The dance teacher ordered a total of 28 outfits and spent $881. If b represents the number of outfits the dance teacher ordered for boys and g represents the number of outfits the dance teacher ordered for girls, which of the following systems of equations can be solved to find b and g ?

Answer

A. $26b + 35g = 28$
$b + g = 881$

B. $26b + 35g = 881$
$b + g = 28$

C. $26g + 35b = 28$
$b + g = 881$

D. $26g + 35b = 881$
$b + g = 28$